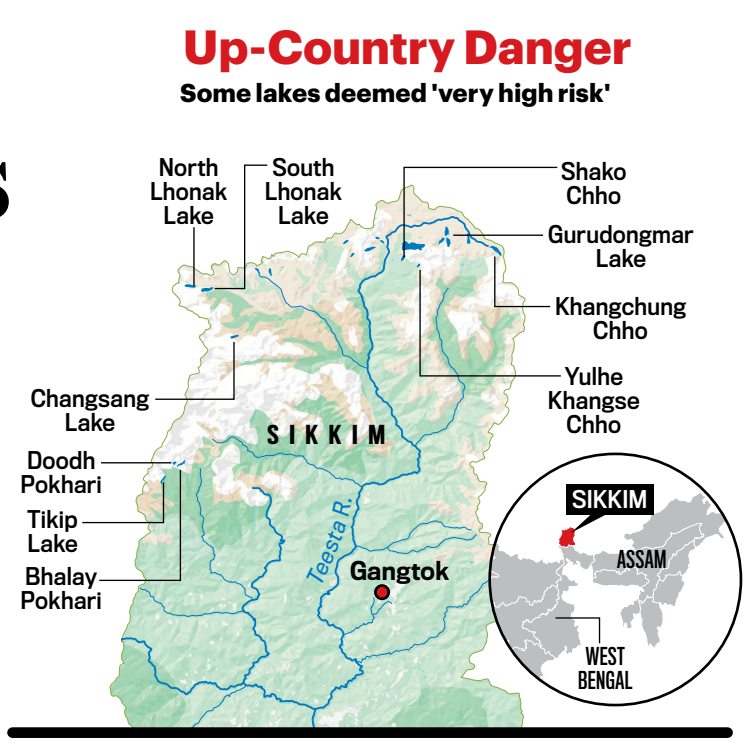
SIKKIM

Mapping the Moods of Lakes

By Arkamoy Datta Majumdar



Nestled amidst the Himalayas, Sikkim has been synonymous with the natural grandeur of its topography and the idyllic, serene beauty of its many mountain lakes. A year and a half ago, the border state faced a calamity that underscored its susceptibility to climate change-induced natural disasters. On October 4, 2023, a cloudburst led the South Lhonak lake, perched at 17,000 feet in the Lhonak Valley, to breach its banks, causing a catastrophic glacial flood that laid entire landscapes to waste and displaced thousands.

The total financial toll is estimated to be a staggering Rs 18,000 crore. The state's most significant investment, the 1,200 MW Sikkim Urja-Teesta Stage III Project, suffered a devastating blow. The ecological impact was equally ruinous: 1,819 hectares of forest cover was lost. The Teesta river, choked with debris, saw its bed level rise by 8-12 metres downstream, triggering frequent floods.

But amid the wreckage, Sikkim has emerged as a beacon of resilience. Spurred by the cataclysmic event, it has launched a mission to develop a homegrown model for mitigating Glacial Lake Outburst Flood (GLOF) risks. It involves undertaking expeditions to remote altitudes to identify high-risk glacial lakes and mapping potential hazards, then designing structural measures to protect against future disasters. With extreme events recurring across the Himalayas, this approach to climate resilience is of universal relevance.

The pioneering effort, a collaborative endeavour among scientists, engineers, environmentalists and local authorities, has yielded a set of proposals that was presented to the Union government on December 23. If adopted, Sikkim's model could pave the way for GLOF risk mitigation across the region, potentially safeguarding millions. "Our mandate was to ensure that we are battle-ready for such incidents. The task was not easy, but we have made significant progress," says Dhirendra Shrestha, principal director of the Sikkim Science and Technology Department, which is spearheading the operations.

Partnering with the National Disaster Management Authority (NDMA), the state assessed its 320 glacial lakes. The NDMA's preliminary assessment, conducted across all Himalayan states, identified 189 high-risk lakes, 40 of which are in Sikkim—the highest density in any state. Of this, 16 were categorised as "very high risk" or Category A lakes.

After this, researchers embarked on multidisciplinary studies. These included hydrometeorological analyses to understand weather-lake interactions, electrical resistivity tomography to map moraine stability, bathymetric investigations (mapping underwater features) to assess lake depth profile, hydrological studies of lake discharge, geological examinations and terrain mapping using UAVs to create 3D models.

With remote assessments alone being insufficient, expedition teams braved the craggy terrain to conduct on-site verifications, trekking to nine of these lakes between September and December 2024. Each expedition comprised about 20 members from the Sikkim government, the Central Water Commission, the Geological Survey of India, NDMA, the Sikkim State Disaster